ADNI — matched CN OOF performance
MAE/RMSE in years; participant-cluster bootstrap 95% CIs

Model	Scans / participants	MAE [95% CI]	RMSE [95% CI]	R^2 [95% CI]	Pearson r [95% CI]
Ridge: DWI-only	131 / 74	7.15 [6.21, 8.12]	8.75 [7.71, 9.77]	-0.721 [-1.496, -0.228]	0.524 [0.375, 0.654]
Ridge: fMRI-only	131 / 74	8.55 [7.54, 9.63]	10.52 [9.31, 11.71]	-1.485 [-2.257, -0.978]	0.270 [0.134, 0.402]
Ridge: Fusion (weighted early)	131 / 74	6.85 [5.92, 7.80]	8.50 [7.48, 9.48]	-0.622 [-1.358, -0.150]	0.544 [0.385, 0.676]
GNN: DWI-only	131 / 74	12.19 [10.74, 13.65]	13.85 [12.32, 15.30]	-3.308 [-5.105, -2.138]	0.396 [0.194, 0.583]
GNN: fMRI-only	131 / 74	9.09 [7.56, 10.72]	11.99 [10.25, 13.67]	-2.229 [-3.359, -1.462]	0.087 [-0.123, 0.273]
GNN: Fusion	131 / 74	8.91 [7.81, 10.06]	10.70 [9.46, 11.85]	-1.572 [-2.362, -1.071]	0.355 [0.193, 0.502]

ADRC — matched CN OOF performance
MAE/RMSE in years; participant-cluster bootstrap 95% CIs

Model	Scans / participants	MAE [95% CI]	RMSE [95% CI]	R^2 [95% CI]	Pearson r [95% CI]
Ridge: DWI-only	243 / 243	6.52 [5.89, 7.25]	8.50 [7.54, 9.48]	0.457 [0.324, 0.568]	0.728 [0.657, 0.788]
Ridge: fMRI-only	243 / 243	8.29 [7.37, 9.18]	11.07 [9.82, 12.18]	0.079 [-0.133, 0.267]	0.514 [0.419, 0.610]
Ridge: Fusion (weighted early)	243 / 243	6.39 [5.74, 7.10]	8.41 [7.44, 9.36]	0.468 [0.339, 0.576]	0.738 [0.673, 0.796]
GNN: DWI-only	243 / 243	8.55 [7.67, 9.50]	11.02 [9.89, 12.15]	0.087 [-0.075, 0.226]	0.412 [0.291, 0.527]
GNN: fMRI-only	243 / 243	9.82 [8.95, 10.75]	12.23 [11.11, 13.38]	-0.125 [-0.324, 0.042]	0.317 [0.192, 0.435]
GNN: Fusion	243 / 243	9.04 [8.15, 10.01]	11.61 [10.49, 12.74]	-0.013 [-0.204, 0.150]	0.401 [0.288, 0.505]

HABS — matched CN OOF performance
MAE/RMSE in years; participant-cluster bootstrap 95% CIs

Model	Scans / participants	MAE [95% CI]	RMSE [95% CI]	R^2 [95% CI]	Pearson r [95% CI]
Ridge: DWI-only	13 / 13	4.12 [2.71, 5.91]	5.06 [3.28, 6.97]	0.722 [0.306, 0.866]	0.873 [0.739, 0.961]
Ridge: fMRI-only	13 / 13	8.06 [3.95, 13.20]	11.75 [5.71, 16.85]	-0.504 [-2.524, 0.330]	0.185 [-0.335, 0.678]
Ridge: Fusion (weighted early)	13 / 13	4.16 [2.02, 7.32]	6.48 [2.72, 10.19]	0.542 [-0.440, 0.925]	0.808 [0.383, 0.975]
GNN: DWI-only	13 / 13	7.14 [4.55, 10.15]	8.87 [5.44, 11.88]	0.143 [-0.916, 0.552]	0.449 [-0.022, 0.865]
GNN: fMRI-only	13 / 13	8.71 [4.04, 14.86]	13.26 [6.18, 19.87]	-0.916 [-2.783, 0.321]	0.094 [-0.467, 0.719]
GNN: Fusion	13 / 13	11.24 [6.42, 17.24]	14.99 [7.65, 21.58]	-1.448 [-3.335, -0.158]	-0.302 [-0.753, 0.413]

Pooled — matched CN OOF performance
MAE/RMSE in years; participant-cluster bootstrap 95% CIs

Model	Scans / participants	MAE [95% CI]	RMSE [95% CI]	R^2 [95% CI]	Pearson r [95% CI]
Ridge: DWI-only	387 / 330	6.66 [6.12, 7.18]	8.49 [7.79, 9.19]	0.620 [0.554, 0.680]	0.788 [0.748, 0.826]
Ridge: fMRI-only	387 / 330	8.37 [7.69, 9.06]	10.91 [9.99, 11.81]	0.374 [0.277, 0.456]	0.618 [0.553, 0.677]
Ridge: Fusion (weighted early)	387 / 330	6.47 [5.92, 7.01]	8.38 [7.66, 9.08]	0.630 [0.566, 0.689]	0.794 [0.753, 0.831]
GNN: DWI-only	387 / 330	9.74 [9.01, 10.48]	11.99 [11.13, 12.82]	0.243 [0.136, 0.332]	0.522 [0.436, 0.603]
GNN: fMRI-only	387 / 330	9.54 [8.74, 10.33]	12.19 [11.23, 13.11]	0.218 [0.127, 0.301]	0.469 [0.382, 0.551]
GNN: Fusion	387 / 330	9.07 [8.37, 9.79]	11.44 [10.58, 12.30]	0.311 [0.226, 0.388]	0.558 [0.485, 0.626]